

Luis A. Rodriguez

Lead Software Engineer & Architect

Skills

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Location: Dallas, TX

Languages	C, JavaScript, TypeScript, CSS, HTML, Java
System & Low Level	Emscripten, WebAssembly, Java Native Interface (JNI), Hardware Emulation, WebSerial, WebUSB, USB Communication, Cross-Compilation, Markdown
Mobile/Web/Frameworks	Angular, Spring Boot, Lua, Gulp, Ant, Mocha, Jasmine, Selenium WebDriver
Leadership	Technical Leadership, Test-Driven Development (TDD), Agile/Scrum, Project Coordination, Git

Work Experience

Texas Instruments -- Software Architect (2010–2025)

- **Calculator Device Emulation:** Designed and implemented the glue layer for targeting embedded code on the web by transpiling to WebAssembly. Coordinated crossfunctional team that did development and test for new features. Designed and implemented an automation entry point using WebSockets.
- **Examcalcs:** Adapted z80 opcodes and emulated the TI-84 Silver Edition and TI-84 Plus C's ASICs. Ported Toshiba T4 cpu emulation from Java and implemented the TI-108 and TI-30MV's ASICs for these involving 800k users went without any reported issues.
- **TI 84 Plus CE ExamCalc** Implemented emulators for z80-based devices whose .
- **Examcalc Accessibiliy:** Implemented code needed in both ez80 assembly and javascript to support market needs to make calculator emulators interact correctly with a screen reader for the vision impaired. Implemented an OCR-type screen translation for scientific and 4-function calculator emulators.
- **UI And Background Thread Refactor:** Worked in conjunction with the development team to refactor the usage of **JNI**, improving performance and memory usage.
- **Open Source Auditing:** Responsible for reviewing software third party library lists to determine their eligibility for integrating into our products.
- **Web Apps Server-Side:** Led team in determining **server configuration** changes needed to address concerns brought up by IT security scans.
- **TI Connect CE:** Delivered cross-platform compatibility for a proprietary CARS communication protocol by porting the codebase to generate platform-native libraries (Windows/macOS). Established the USB communication architecture using an abstraction layer and integrated the functionality into the application via Java Native Interface (JNI).

Digital Chocolate -- Game Engineering Manager (2009–2010)

- Coordinated project planning for various US carriers, optimizing production costs and improving revenue in collaboration with sales and marketing teams.
- Led and mentored the engineering team, providing tutorials and code deep-dives.
- **Brew In App Purchase Library (2010):** Created module for Qualcomm's Application Value Billing API (C/C++), promoting code reuse across future game titles.
- **Rollercoaster Rush for iPhone (2010):** Ported the OpenGL game implementation from iPhone to **Brew**.

LemonQuest -- iPhone Lead Programmer (2008–2009)

- **Circulate and Circulate Prologue (2008):** Ported a PC-based DirectX game to the iPhone's **OpenGL ES**, translating levels for a touch interface. The game earned a **Bronze Medal** from PocketGamer.
- **iFit (2009):** Developed a fitness application for the iPhone with pedometer functionality and fitness tips.

Gameloft -- 3d/iPhone Programmer (2005–2008)

- **Blockbreaker for iPhone (2007–2008):** Developed and ported the first successful prototype for the brickbreaker game using the Celestial framework, leading to the studio being assigned more iPhone projects.
- **3D Programmer (2006–2007):** In charge of porting multiple 3D game titles to resource-limited, 3D-capable devices using **Java/J2ME**.

Education

Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE)

- **Completed coursework towards:** Master's in Electronics and Telecommunications (Specialization in Control Systems and Automation), 2002–2005

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM)

- **Bachelor's in Electronics Systems Engineering**, 1997–2001

Online Courses

- **JavaScript & Web:** JavaScript Design Patterns, Object-Oriented JavaScript, JavaScript Basics, Intro to HTML and CSS, HTML5 Game Development, jQuery (2013-2015)
- **Other:** Website Performance Optimization (2014), Software Testing (2012), Web Application Engineering (2012), Artificial Intelligence for Robotics, Introduction to Artificial Intelligence (2011)
- **iOS Development:** iOS Persistence and Core Data, iOS Networking with Swift, UIKit Fundamentals I & II, Intro to iOS App Development Swift (2015)